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**Question 1:**

**Answer:**

Number of full-duplex channels = Total bandwidth / (2 × bandwidth per simplex channel)

= 45 MHz / (2 × 25 kHz)

= 45,000 kHz / 50 kHz

= 900

**Question 2:**

**Answer:**

Channels per cell = Total channels / Cluster size

= 560 / 7

= 80 channels per cell.

**Question 3:**

**Answer:**

Total simultaneous voice channels = Number of carriers × Time slots per carrier

= 100 × 8

= 800 simultaneous voice channels.

**Question 4:**

**Answer:**

Power ratio (dB) = $10 \log_{10}$(Base station power / Handset power)

= $10 \log_{10}(80 / 4)$

= $10 \log_{10}(20)$

= 13.01 dB.

**Question 5:**

**Answer:**

C/I (dB) = Carrier power (dBm) − Interference power (dBm)

= −60 − (−92)

= 32 dB.

**Question 6;**

**Answer:**

Total channels = 42 MHz / 300 kHz = 42,000 / 300 = 140 channels.

Channels per cell = 140 / 7 = 20 channels per cell.

**Question 7:**

**Answer:**

Received power varies inversely with distance raised to the path-loss exponent.

$P_2 = P_1 \times (d_1 / d_2)^n$

$= 64 \times (1 / 2)^4$

= 64 / 16

= 4 W.

**Question 8:**

**Answer:**

Transmit power factor = (New radius / Original radius)²

= (2R / R)²

= 2²

= 4.

**Question 9:**

**Answer:**

Total channels in cluster = Cluster size × Channels per cell

= 4 × 15

= 60 channels.

**Question 10:**

**Answer:**

Convert speed: v = 100 × 1000 / 3600 = 27.78 m/s.

Maximum Doppler shift: $f^{d} = (v / c) \times f^{c}$

$= (27.78 / (3 \times 10^{8})) \times 900 \times 10^{6}$

≈ 83.33 Hz.

**Question 11:**

**Answer:**

Time-slot duration = Frame duration / Number of slots

= 5 ms / 8

= 0.625 ms.

**Question 12:**

**Answer:**

Number of carrier frequencies = Total bandwidth / Carrier spacing

= 24 MHz / 200 kHz

= 24,000 kHz / 200 kHz

= 120 carrier frequencies.

**Question 13:**

**Answer:**

Convert transmit power to dBm:

$P_t = 10 \log_{10}(8 \times 1000) = 39.03$ dBm.

Received power = Transmit power − Path loss

= 39.03 − 33

= 6.03 dBm.

**Question 14:**

**Answer:**

Using frequency in MHz and distance in km:

$FSPL = 32.44 + 20 \log_{10}(f) + 20 \log_{10}(d)$

$= 32.44 + 20 \log_{10}(1900) + 20 \log_{10}(2.5)$

= 105.98 dB.

**Question 15:**

**Answer:**

Total capacity = Number of cells × Calls per cell

= 6 × 250

= 1,500 simultaneous calls.

**Question 16:**

**Answer:**

Average call duration = 3 × 60 = 180 seconds.

Using N = Call arrival rate × Average call duration:

N = 4000 × 180 = 720,000 active calls.

**Question 17:**

**Answer:**

Total bandwidth = Number of channels × Bandwidth per channel

= 150 × 25 kHz

= 3,750 kHz

= 3.75 MHz.

**Question 18:**

**Answer:**

For 30 adjacent bands, the number of gaps is 30 − 1 = 29.

Total guard-band bandwidth = 29 × 40 kHz

= 1,160 kHz

= 1.16 MHz.

**Question 19:**

**Answer:**

Cell diameter = 2 × 1.5 = 3 km.

Handoffs per hour ≈ Distance travelled per hour / Cell diameter

= 90 km / 3 km

≈ 30 handoffs per hour.

**Question 20:**

**Answer:**

Total base stations = 8 × 40 = 320.

Total user capacity = 320 × 250 = 80,000 users.

**Question 21:**

**Answer:**

Convert radius: 400 m = 0.4 km.

Approximate cell area = $R^2 = 0.4^2 = 0.16$ km².

Number of cells = 400 / 0.16 = 2,500 cells.

**Question 22:**

**Answer:**

Convert total bandwidth: 6 MHz = 6,000 kHz.

Number of channels = 6,000 / 30 = 200 channels.

**Question 23:**

**Answer:**

Traffic per subscriber = (Calls per hour × Average duration in minutes) / 60

= (4 × 2.5) / 60 = 0.1667 Erlangs.

Total offered traffic = 1,200 × 0.1667 ≈ 200 Erlangs.

**Question 24:**

**Answer:**

Number of simultaneous users = Total bandwidth / Bandwidth per user

= 40 MHz / 400 kHz

= 40,000 kHz / 400 kHz

= 100 users.

**Question 25:**

**Answer:**

Total user capacity = Number of base stations × Users per base station

= 1,200 × 600

= 720,000 users.